EMIRHAN YAMAN

Backend-Focused Full-Stack Software Engineer

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Software engineer building production applications with .NET, EF Core and SQL, with full-stack experience across Next.js, React, TypeScript and Blazor.

PROFESSIONAL EXPERIENCE

Tekhnelogos | Enerwise - Freelance Software Engineer — Oct 2024 - Present

- Contributed to Enerwise energy management from its early stages: designed databases, developed .NET/C# APIs and background workers, and built React/TypeScript features with Redux and React Query.
- Built CBAM compliance and GRI sustainability reporting features for the Enerwise carbon management platform using .NET, EF Core, SQL, Blazor and MudBlazor.
- Optimized reporting queries for large emissions datasets using database indexing, EF Core AsNoTracking and projections that fetch only required fields, improving report retrieval and application responsiveness.
- Integrated SharePoint document management with Microsoft Graph and Azure AD (Entra ID), supporting large file uploads, secure sharing and file operations.

Asis Elektronik ve Bilişim Sistemleri A.Ş. - Software Engineering Intern — Jul - Aug 2024

- Analyzed .NET projects using DDD, CQRS and Mediator; gained hands-on experience with RabbitMQ, gRPC/REST APIs, Entity Framework/LINQ and Docker.

TECHNICAL SKILLS

Backend: C#, .NET / ASP.NET Core, EF Core, REST APIs, MediatR
Frontend: Next.js, React, Blazor, TypeScript, MudBlazor, Redux, TanStack Query
Messaging / Real-Time: RabbitMQ, SignalR, Socket.IO
Data: SQL, PostgreSQL, SQL Server, MongoDB, Redis
Cloud / Integrations: Docker, AWS S3, Azure AD / Entra ID, Vercel Blob
AI / Tool Integration: Model Context Protocol (MCP), Ollama, RAG, pgvector

PROJECTS

CommercePilot - MCP-Powered AI Shopping Assistant | GitHub

- Built a .NET 10 Model Context Protocol (MCP) server and client with the official C# SDK and Streamable HTTP, exposing catalog, cart and order workflows through validated tools.
- Connected local LLMs to MCP tools with scoped authorization; added OAuth/PKCE for external clients and required explicit user approval for checkout, cancellation and refunds.
- Built Next.js/React streaming chat and retrieval-augmented generation (RAG) with Ollama embeddings and PostgreSQL/pgvector, returning source-linked answers from versioned site knowledge.

InventoryManagement - Multi-Company Inventory Platform | GitHub / Live Demo

- Built a multi-company inventory system with .NET 10, EF Core, MediatR/CQRS, Next.js/TypeScript and PostgreSQL; implemented company-scoped access, role-based permissions, stock transfers and purchasing workflows.
- Integrated Keycloak authentication and SignalR updates; deployed Docker services to Oracle Cloud (ARM64) with GitHub Actions CI/CD, separate staging/production environments and Nginx/HTTPS.

WorkGrid - Task Management Platform | Server / Client

- Built workspace, board and task management with a .NET 9/EF Core/MediatR backend and Next.js/TypeScript frontend, using React Query for server state and AWS S3 for attachments.
- Implemented RabbitMQ-driven background services for asynchronous activity logging and notification processing, with SignalR for real-time delivery.

AI-Powered Food Ingredient Detection & Recipe Recommendation System

YOLO · Faster R-CNN · Python · FastAPI · React Native/Expo · Node.js/Express · MongoDB

- Benchmarked 15 YOLO and Faster R-CNN variants across 45 training runs on a custom 36-class dataset (1,901 images; 4,761 annotations), using repeated-seed and class-wise evaluation.
- Built a mobile system with server-side YOLOv9m inference, recipe ranking, missing-ingredient detection, shopping lists and nearby-store mapping using demonstration store, inventory and price data.

MyMedia - Full-Stack Social Media Platform | Server / Client

- Built a layered .NET 9 Web API with EF Core, SQL Server, MediatR/CQRS, FluentValidation, JWT access/refresh tokens and centralized exception handling.
- Implemented posts, media uploads, likes, saved posts, threaded comments, profiles and public/private follow workflows; built the Next.js/TypeScript frontend with TanStack Query, Zustand and Vercel Blob.

Virtual Reality Organ Simulation Application (TÜBİTAK-2209)

- Developed interactive, audio-supported anatomy experiences for elementary and middle school students under TÜBİTAK-2209, using Unity, C#, VR, Oculus Quest and Blender.

EDUCATION

Bachelor of Computer Engineering
Bozok University, Yozgat | 2020 - 2024

LANGUAGES

Turkish (Native) | English (B2)
Spanish (A2)

REFERENCES

Mustafa Uçak | General Manager, Enerwiseai | mustafa.ucak@yapayzekaajansi.com
Yasin Kurt | Software Team Lead, Enerwiseai | yasin.kurt@enerwiseai.com